



Optimal photosynthetic strategies

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Outline

- Eco-evolutionary optimality theory
- Global optimal photosynthetic traits
- Site-level trait variability
- Implications

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Eco-evolutionary optimality theory

“EEO invokes the **power of natural selection to eliminate uncompetitive trait combinations**, and thereby shape predictable, general patterns in vegetation”

Tansley review | [Free Access](#)

Eco-evolutionary optimality as a means to improve vegetation and land-surface models

Sandy P. Harrison , Wolfgang Cramer, Oskar Franklin, Iain Colin Prentice, Han Wang, Åke Brännström, Hugo de Boer, Ulf Dieckmann, Jaideep Joshi, Trevor F. Keenan, Aliénor Lavergne, Stefano Manzoni, Giulia Mengoli, Catherine Morfopoulos, Josep Peñuelas, Stephan Pietsch, Karin T. Rebel, Youngryel Ryu, Nicholas G. Smith, Benjamin D. Stocker, Ian J. Wright ... [See fewer authors](#) ^

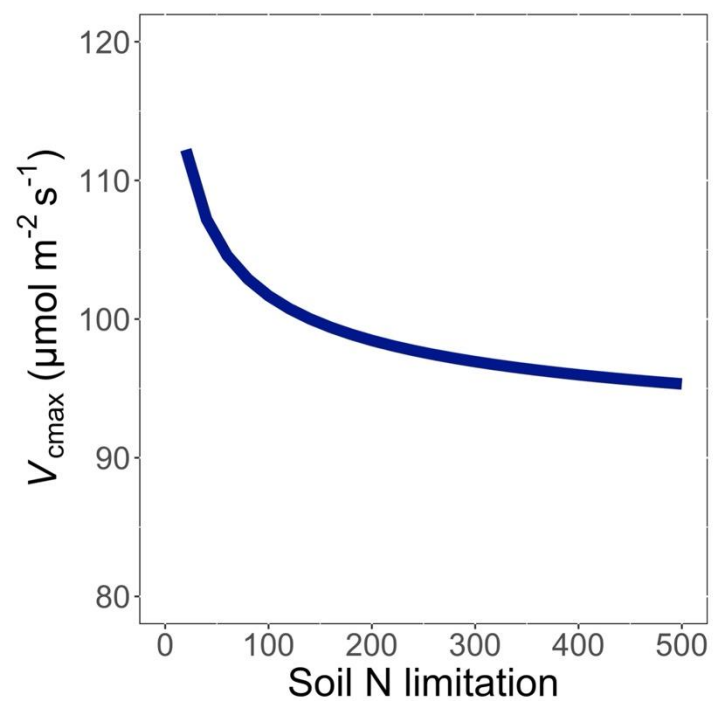
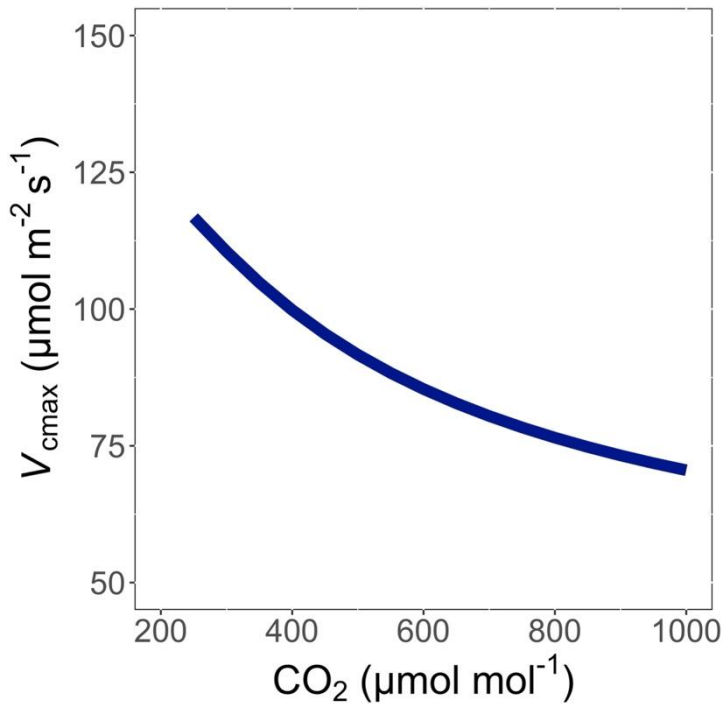
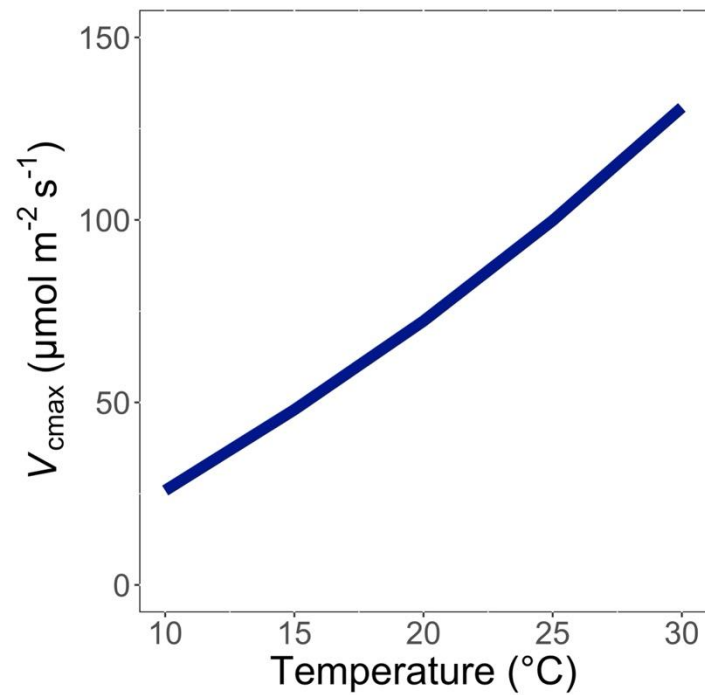
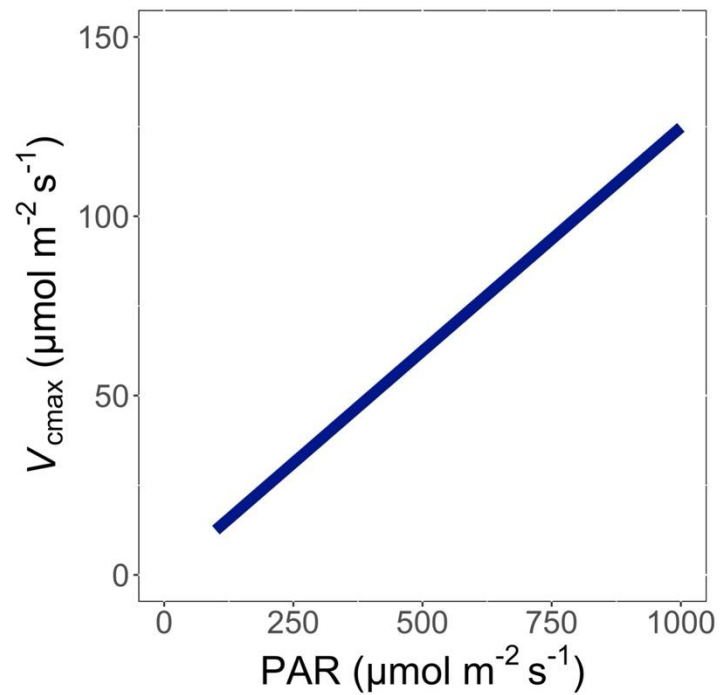
First published: 15 June 2021 | <https://doi.org/10.1111/nph.17558>

Photosynthesis optimization: least cost

Optimally, plants will maintain fastest rate of photosynthesis at the lowest summed resource cost (water and nutrient use)

Optimal trait example

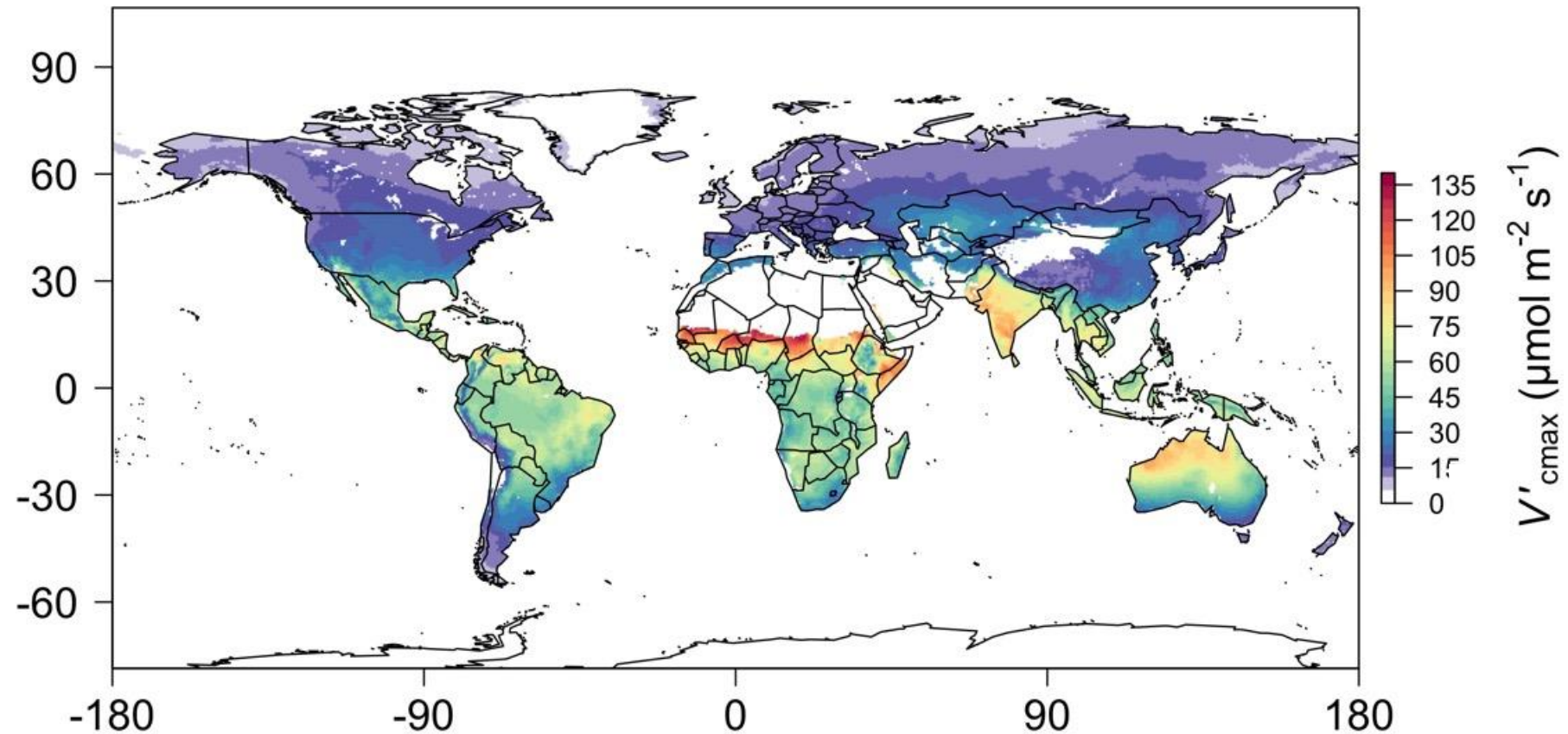
The maximum rate of Rubisco
carboxylation (V_{cmax})



Trait-environment relationships can be predicted in ways that are:

1. Parameter-sparse
2. Mechanistically interpretable

We can predict optimal traits in different environments

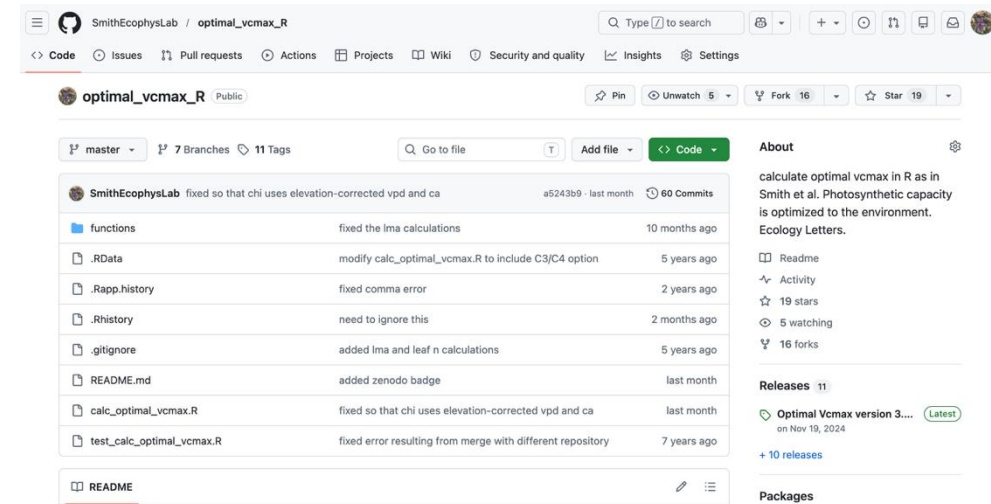


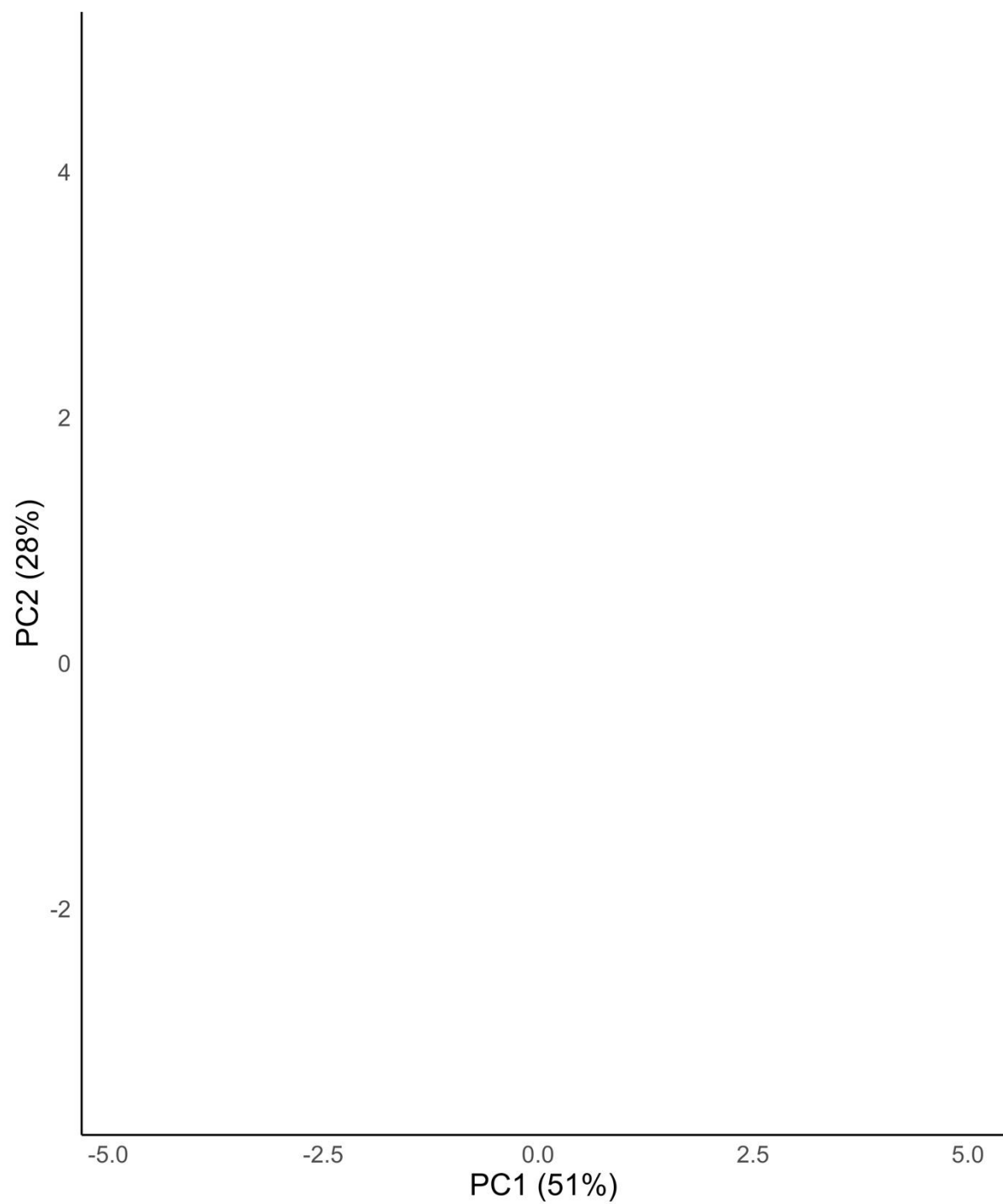
Outline

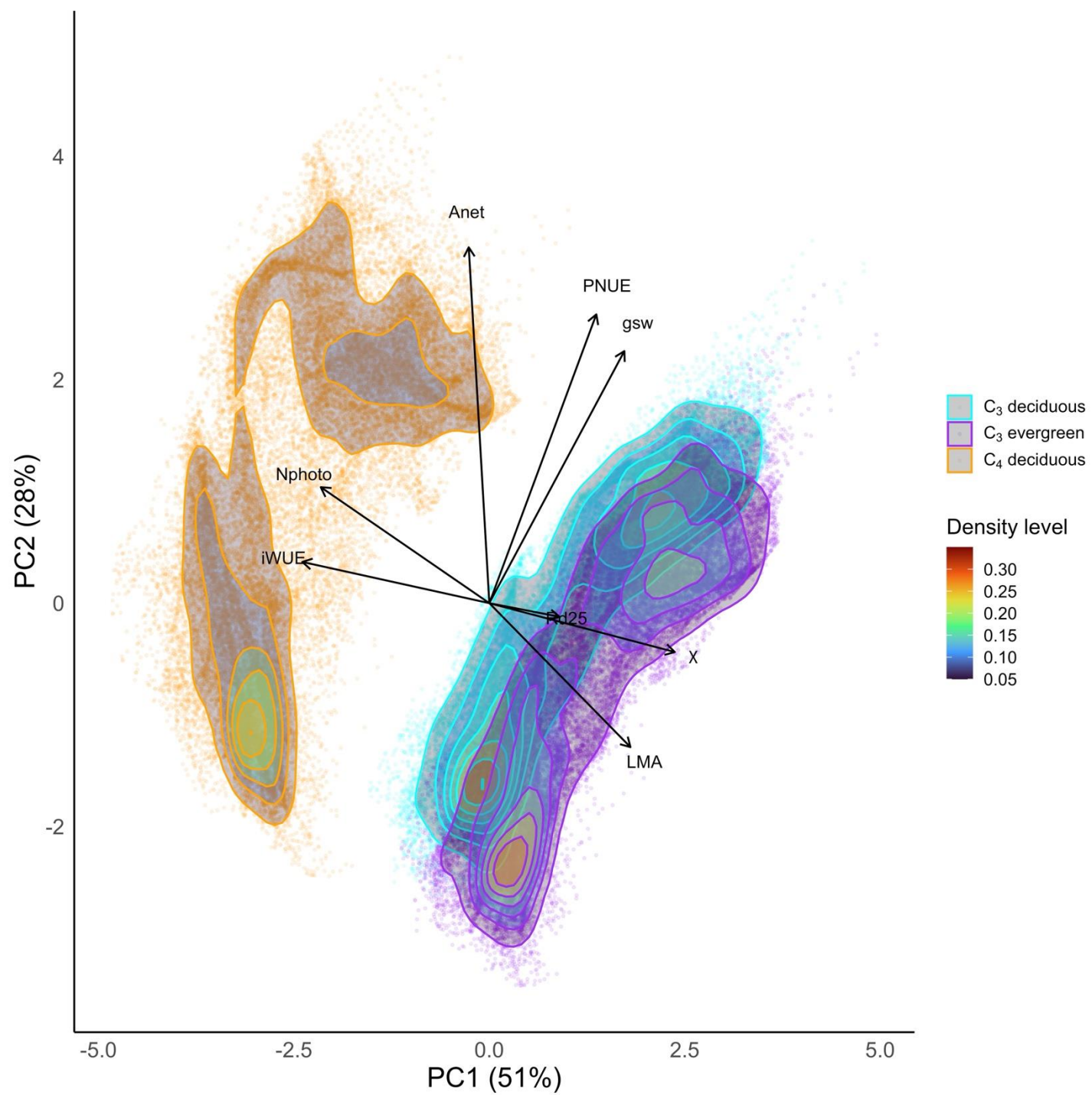
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Global simulation strategy

1. Developed optimal photosynthesis model
2. Used globally gridded climate to simulate traits
3. Used PCA to explore trait relationships and clustering
 - Carbon uptake (A_{net})
 - Carbon loss (R_d)
 - Water use for photosynthesis (g_{sw})
 - Nitrogen use for photosynthesis (N_{photo})
 - Leaf economics (LMA)
 - Least-cost theory (χ)
 - Photosynthetic efficiency ($iWUE$, $PNUE$)

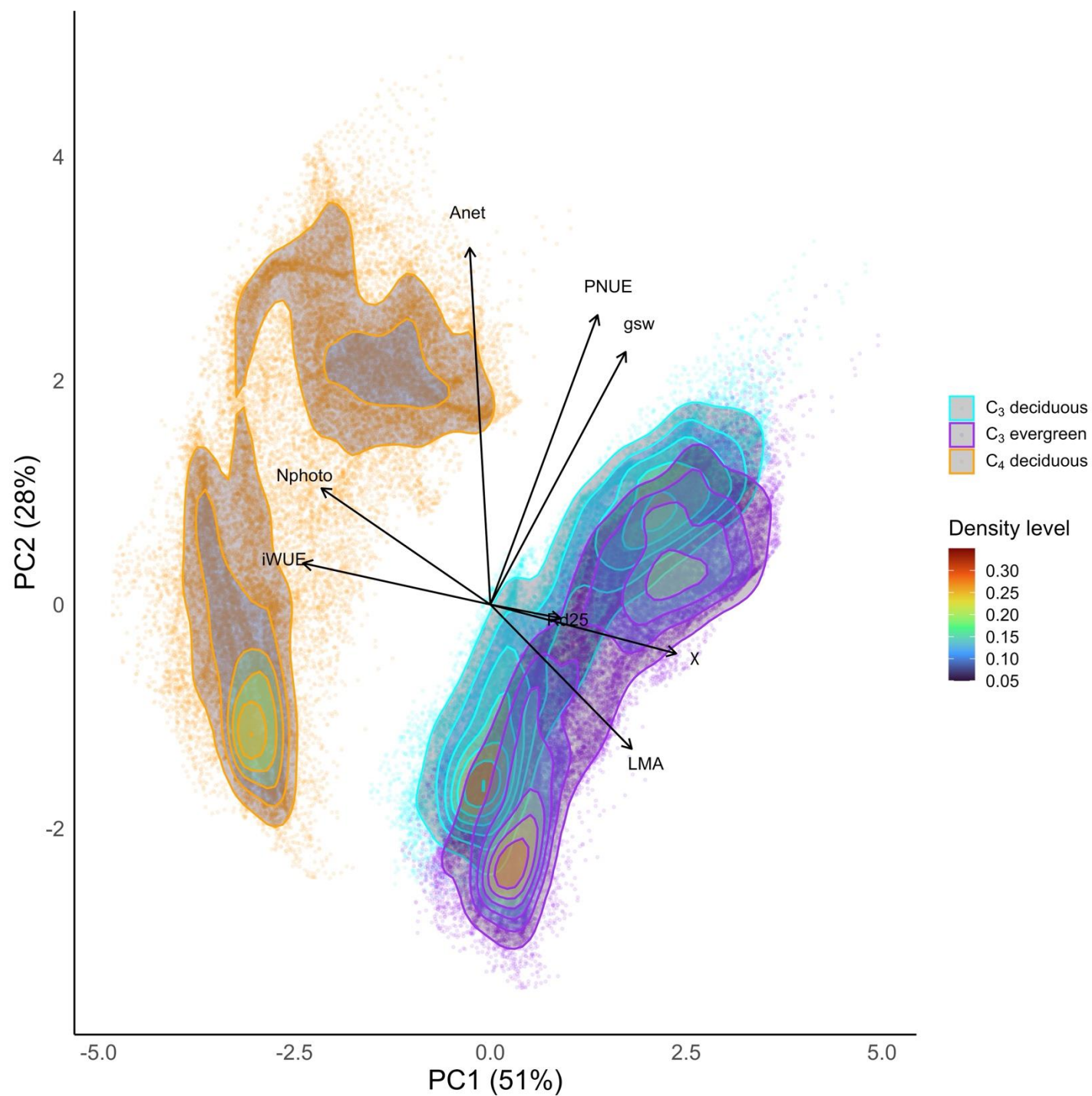






PC1

- N use-WUE
- C₃-C₄ divide

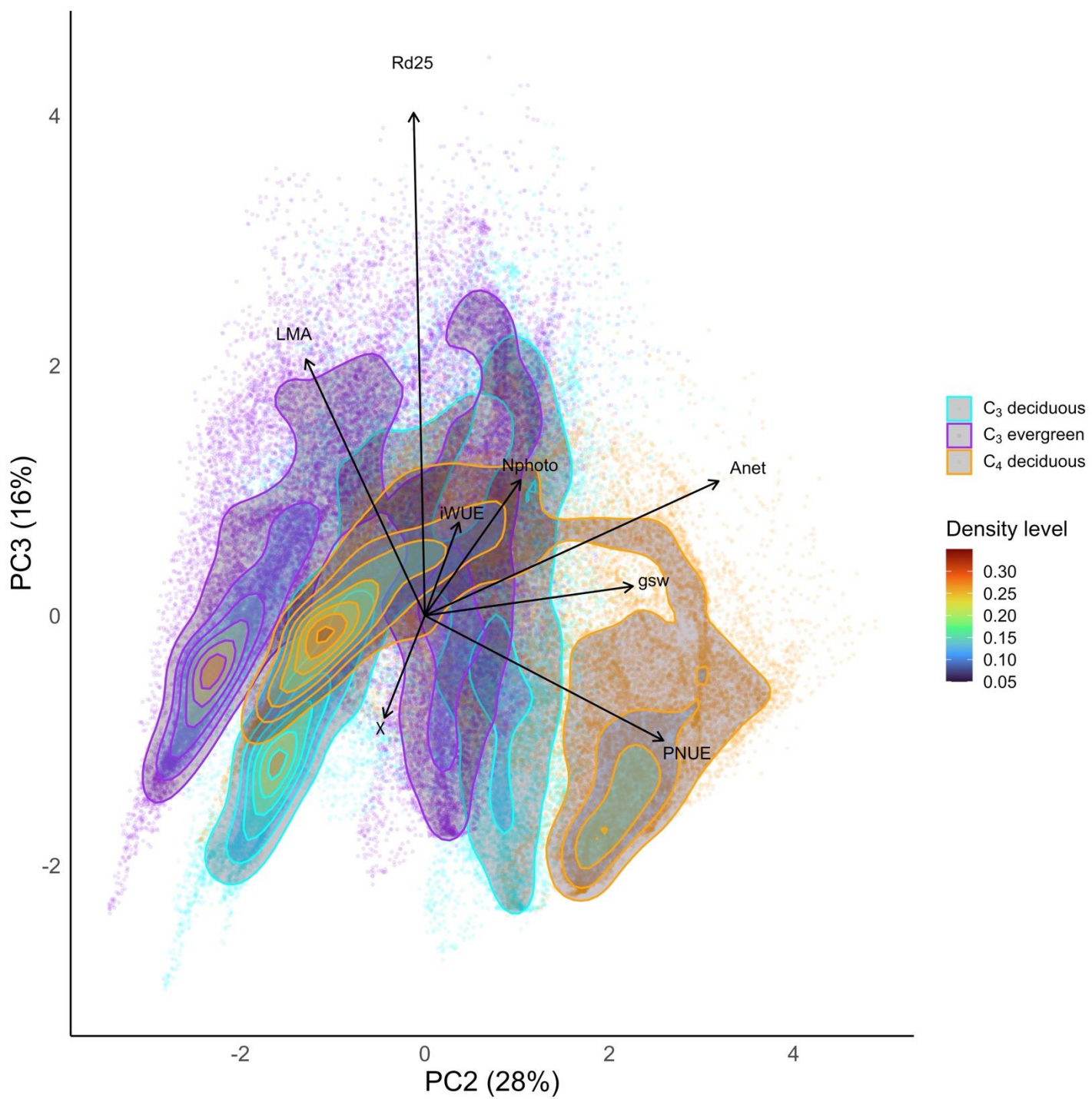


PC1

- N use-WUE
- C₃-C₄ divide

PC2

- H₂O use-PNUE
- A_{net}
- wet-dry divide



PC3

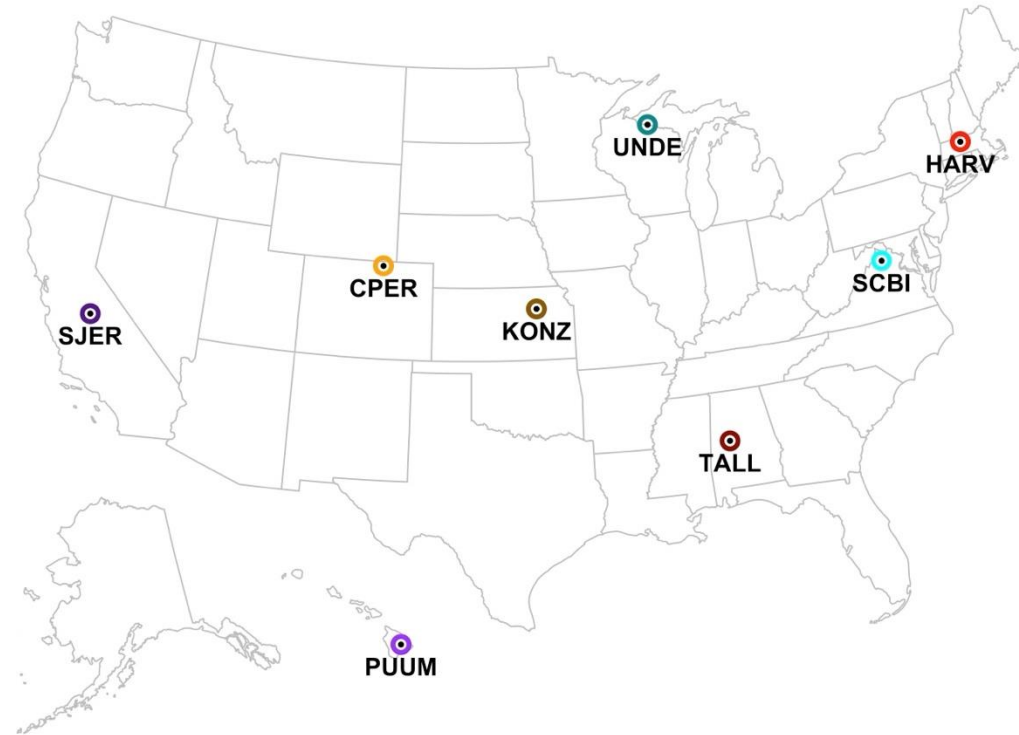
- LMA, R_d
- Leaf carbon costs

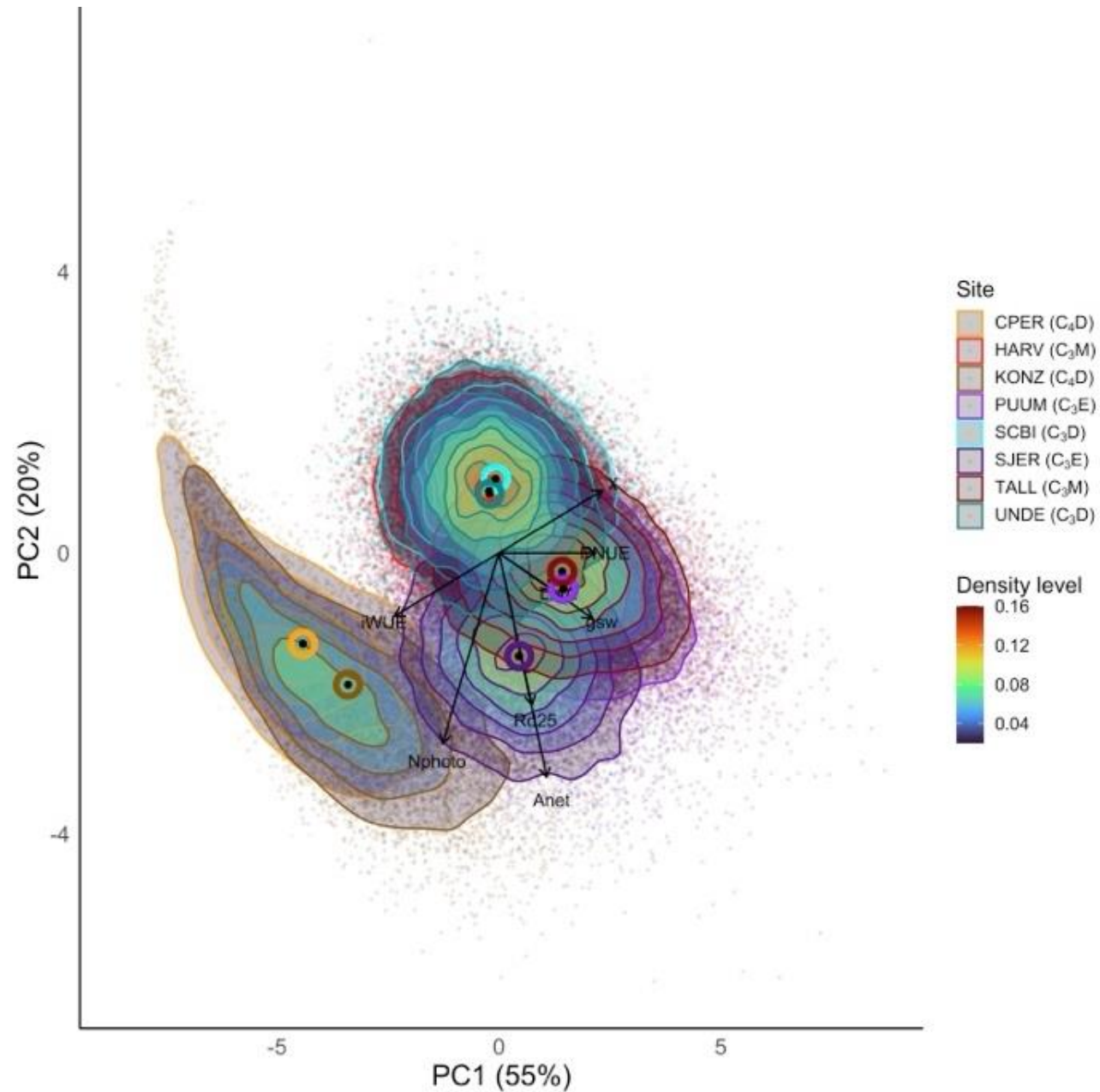
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Site-level simulation strategy

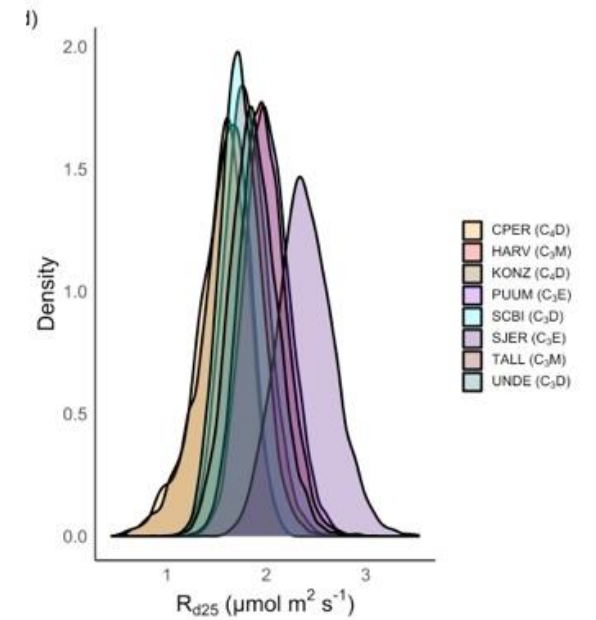
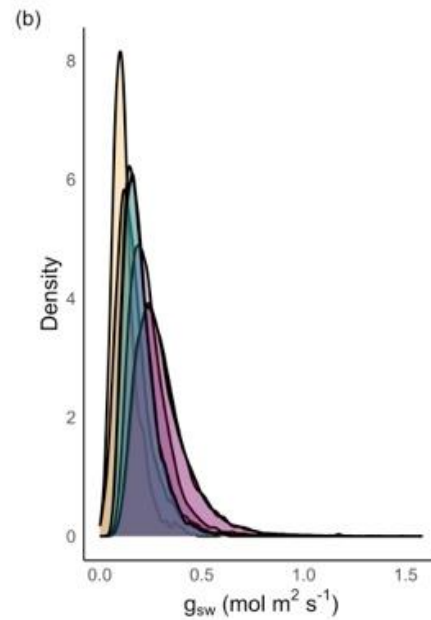
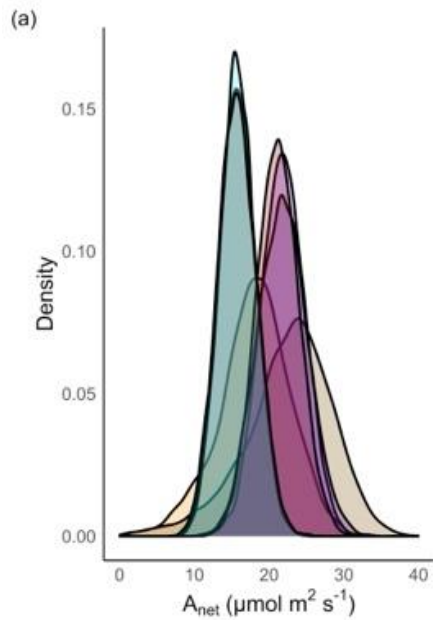
- Chose sites representing different dominant PFTs
 - NEON network
 - Deciduous trees, evergreen trees, mixed forests, C₄ grasses
- Simulated optimal traits 10,000x using climate distribution (mean $\pm$ 0.1*mean)



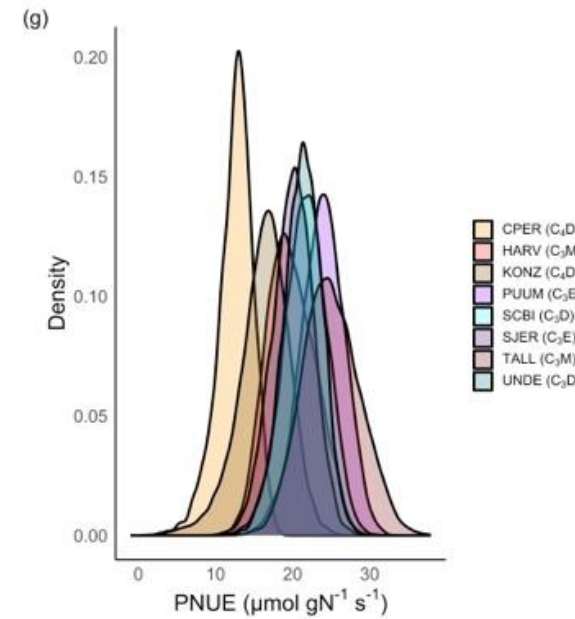


PCs defined similarly
to global simulation

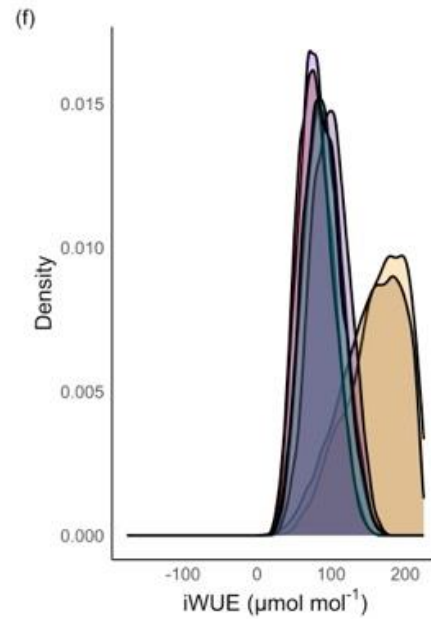
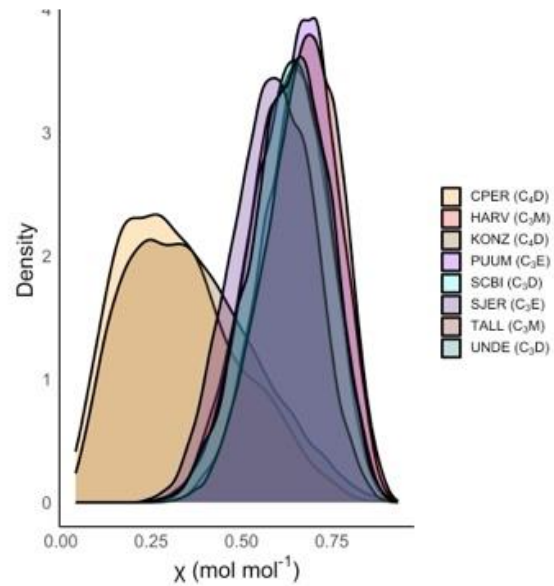
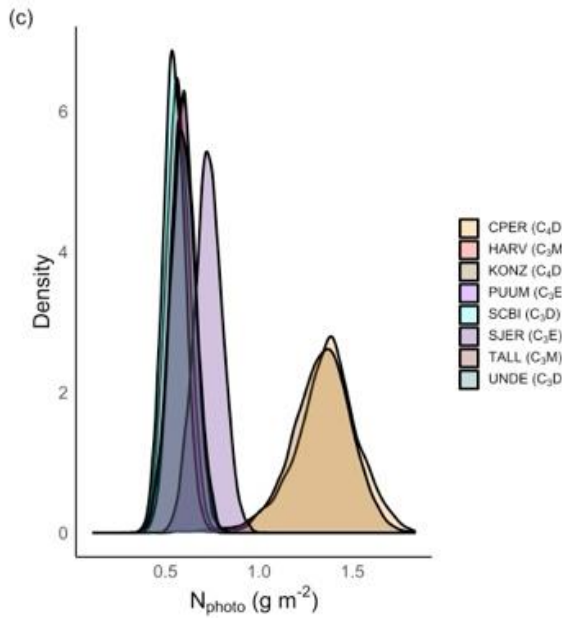
A lot of within-site
variability



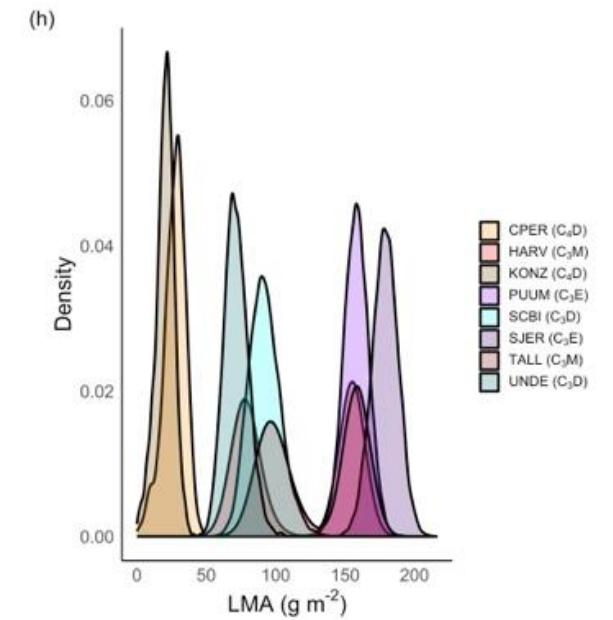
Within-site variability is larger than across-site for many traits



Some traits show
separation by
photosynthetic type



LMA shows separation by
PFT



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- The three defining axes of optimal photosynthetic strategies:
 1. Nitrogen use-water use efficiency (PFT-driven)
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 3. Tissue costs of photosynthesis

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- **Plant type** dictates optimal photosynthetic strategies via some (LMA, WUE, N use), but not all traits

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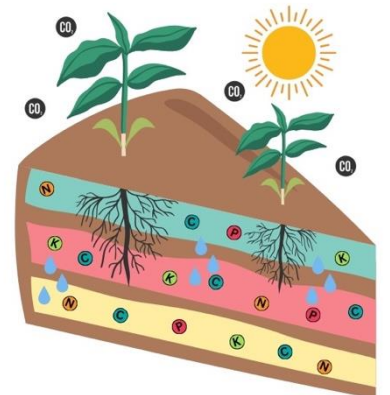
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- Plant type dictates optimal photosynthetic strategies via some (LMA, WUE, N use), but not all traits
- There are **many optimal strategies** within an environment

Quick notice

- Lab is moving to the University of Nebraska - Lincoln
- Starting Fall 2026
- Reach out if you are interested in collaborating



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LINCOLN



Presentation: github.com/SmithEcophysLab/esa_2026

Model code: github.com/SmithEcophysLab/optimal_vcmax_r

Contact: nsmith87@unl.edu



